System of units

$$\hbar = c = 1$$

Dimensions:

$$[\text{Energy}] = [\text{mass}] = [\text{length}]^{-1} = [\text{time}]^{-1}$$

Relationships:

— Energy: $E^2 = \mathbf{p^2} + \mathbf{m^2}$, $E = \omega$

— Wavelength: $\lambda = \frac{1}{p}$

— Time: $t = \frac{1}{E}$

Conversions:

$$\hbar c = 197.327 \text{ MeV fm}$$

$$1 \text{ GeV}^{-1} = 0.197 \text{ fm}$$

$$1 \text{ fm} = 5.07 \text{ GeV}^{-1}$$

$$1 \text{ s} = 1.52 \times 10^{24} \text{ GeV}^{-1}$$

1. Matter